

SE LAB 3 - UML COMPONENT DIAGRAM

Pharmacy Expiry & Re-order Dispatch Engine

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Architecture Selection

1. Architectural Choice

Layered Architecture for the Pharmacy Expiry & Re-order Dispatch Engine. The system is organized into separate layers for user interaction, pharmacy operations, and data storage. The Pharmacy Staff User Interface handles interaction with the system, while Inventory & Batch Management, Expiry Monitoring & Alerts, and Automated Re-order Management handle the main pharmacy operations. The Pharmacy Database stores the required medicine, batch, and order information.

2. Reason 1 – Clear Separation of Responsibilities

The architecture separates the different responsibilities of the pharmacy system. The Pharmacy Staff User Interface is responsible for user interaction, Inventory & Batch Management handles inventory-related operations, Expiry Monitoring & Alerts handles expiry checking and alerts, and Automated Re-order Management handles re-order processing. The Pharmacy Database is kept separate for storing system data. This makes the system easier to understand and maintain.

3. Reason 2 – Easier Maintenance and Modification

The separation of components makes it easier to modify individual functionalities without affecting the entire system. For example, changes to expiry monitoring can be made within the Expiry Monitoring & Alerts component, while changes to re-order processing can be handled within Automated Re-order Management. This allows the system to be updated more easily as pharmacy requirements change.

4. Security Advantage

The layered structure provides a clear boundary between the Pharmacy Staff User Interface and the Pharmacy Database. Pharmacy staff access inventory and batch information through the application components rather than directly accessing the database. This allows authorization checks to be applied before inventory information is modified and helps protect pharmacy data from unauthorized access.

5. Performance Benefit

Separating the pharmacy operations into dedicated components allows frequently used operations such as inventory access, expiry monitoring, and re-order processing to be handled independently. This avoids putting unnecessary processing in the user interface and allows the system to efficiently retrieve and update information through the appropriate components.

UML COMPONENT DIAGRAM :

